

12

RATIO AND PROPORTION

Regularly (1-2) questions have been asked from this chapter. Generally both the topics i.e. ratio and proportion have same importance but more emphasis is given on proportion. The concepts of this chapter are extremely used in other chapters like-age, average. etc.



RATIO

A ratio is the comparison of two or more quantities of the same type (or kind) by division. i.e. if a and b are two quantities of same kind (or same unit), then the fraction a/b is called the ratio a to b and we write it as $a : b$. Here, a is called antecedent and b is called the consequent.

e.g. If a fruit box contains 8 oranges and 7 lemons then the ratio of oranges to lemons is 8 to 7 or 8:7.

Note If both terms a and b of a ratio are multiplied or divided by the same quantity, then ratio remains unchanged.

i.e. $a:b$ is same as $na:nb \Rightarrow \frac{a}{b} = \frac{na}{nb}$ and $a:b$ is same as $\frac{a}{n}:\frac{b}{n} \Rightarrow \frac{a}{b} = \frac{a/n}{b/n}$.

EXAMPLE 1. If the ratio of 90 cm to 1.5 m is same as 3 : x , then x is

a. 4

b. 5

c. 2

d. 1

Sol. b. 1.5 m = 150 cm (units must be same)

$$\text{Ratio of 90 cm to 1.5 m} = \frac{90}{150} = \frac{90/30}{150/30} = \frac{3}{5}.$$

As this ratio is same as 3 : x

$$\therefore \frac{3}{x} = \frac{3}{5}$$

$$\Rightarrow 3x = 5 \times 3$$

$$\Rightarrow x = 5$$

Types of Ratios

1. **Compound Ratio** When two or more ratios are multiplied together, they are said to be compound ratio.

If $\frac{a}{b}$, $\frac{c}{d}$, $\frac{e}{f}$ and $\frac{g}{h}$ are all ratios, then their compound ratio is $aceg : bdfh$.

e.g. Compound ratio of $\frac{2}{3}$, $\frac{4}{7}$ and $\frac{1}{3} = \frac{2 \times 4 \times 1}{3 \times 7 \times 3} = \frac{8}{63}$.

2. **Duplicate Ratio** When a ratio is compounded with itself, the resulting ratio is called the duplicate ratio.

So, $a^2 : b^2$ is duplicate ratio of $a : b$.

EXAMPLE 2. The duplicate ratio of the ratio $2\sqrt{2} : 3\sqrt{5}$ is

- a. 4 : 9 b. 8 : 45 c. 2 : 3 d. 6 : 45

Sol. b. The duplicate ratio of $2\sqrt{2} : 3\sqrt{5}$ is

$$\frac{2\sqrt{2}}{3\sqrt{5}} \times \frac{2\sqrt{2}}{3\sqrt{5}} = \frac{4\sqrt{4}}{9\sqrt{25}} = \frac{8}{45} = 8 : 45$$

3. **TriPLICATE Ratio** If a ratio is compounded three times with itself, then resulting ratio is called triplicate ratio.

So, $a^3 : b^3$ is the triplicate ratio of $a : b$.

EXAMPLE 3. If $4x + 3 : 9x + 10$ is the triplicate ratio of $3 : 4$. Then, the value of x is

- a. 2 b. 6 c. 8 d. 12

Sol. b. Since, $4x + 3 : 9x + 10$ is the triplicate ratio of $3 : 4$.

$$\text{Therefore, } \frac{4x+3}{9x+10} = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \Rightarrow \frac{4x+3}{9x+10} = \frac{27}{64}$$

$$\Rightarrow 64(4x+3) = 27(9x+10)$$

$$\Rightarrow 256x + 192 = 243x + 270$$

$$\Rightarrow 256x - 243x = 270 - 192 \Rightarrow 13x = 78$$

$$\therefore x = \frac{78}{13} = 6$$

Hence, the value of x is 6.

4. **Subduplicate Ratio** If two numbers are in ratio, then the ratio of their square roots is called subduplicate ratio.

If $a : b$ is ratio, then the subduplicate ratio is $\sqrt{a} : \sqrt{b}$.

e.g. Subduplicate ratio of $9 : 16$ is $\sqrt{9} : \sqrt{16} = 3 : 4$.

5. **Subtriplicate Ratio** If two numbers are in ratio, then the ratio of their cube roots is called subtriplicate ratio.

If $a : b$ is ratio, then the subtriplicate ratio is $\sqrt[3]{a} : \sqrt[3]{b}$.

e.g. Subtriplicate ratio of $64 : 27$ is $\sqrt[3]{64} : \sqrt[3]{27} = 4 : 3$

6. **Reciprocal Ratio** If $a : b$ is a ratio, then $\frac{1}{a} : \frac{1}{b}$ is its reciprocal ratio, i.e. $b : a$.

e.g. Reciprocal ratio of $3 : 7$ is $\frac{1}{3} : \frac{1}{7}$ i.e. $7 : 3$.

EXAMPLE 4. Compound ratio of the duplicate ratio of $5 : 6$, the reciprocal ratio of $25 : 42$ and the subduplicate ratio of $36 : 49$ is

- a. 1 : 2 b. 2 : 3 c. 1 : 1 d. 4 : 9

Sol. c. The duplicate ratio of $5 : 6$ is $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

The reciprocal ratio of $25 : 42$ is $\frac{1}{25} : \frac{1}{42}$ i.e. $\frac{42}{25}$

and subduplicate ratio of $\frac{36}{49}$ is $\frac{\sqrt{36}}{\sqrt{49}} = \frac{6}{7}$

$\therefore$ Compound ratio is $\frac{25}{36} \times \frac{42}{25} \times \frac{6}{7} = \frac{1}{1} = 1 : 1$

IMPORTANT POINTS

- Usually, the ratio is expressed in its lowest terms.
- Ratio exists only between quantities of the same kind.
- Ratio is a fraction, so it has no units.
- Ratio is taken only between positive quantities.

PROPORTION

The equality of two ratios is called proportion.

' $:$ ' is the sign of proportion.

So, if $a : b = c : d$, we write it as

$a : b :: c : d$ and say that a, b, c and d are in proportion.

The quantities a, b, c and d are called the first, second, third and fourth terms of proportion respectively. First and fourth terms are called **extreme terms** and second and third terms are called **means** or **middle terms**.

If quantities a, b, c and d are in proportion, then

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

i.e. product of extreme terms = product of middle terms.

This is also called cross-product rule.

Types of Proportion

1. **Continued Proportion** The non-zero quantities of same kind, $a, b, c, d, e, f, \dots$ are said to be continued proportion, if $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} = \dots$

EXAMPLE 5. If a, b, c, d and e are in continued proportion, then a/e is equal to

- a. a^3/b^3 b. a^4/b^4 c. b^3/a^3 d. b^4/a^4

Sol. b. Since, a, b, c, d and e are in continued proportion.

$$\begin{aligned} \therefore \frac{a}{b} &= \frac{b}{c} = \frac{c}{d} = \frac{d}{e} \Rightarrow \frac{e}{d} = \frac{d}{c} = \frac{c}{b} = \frac{b}{a} \Rightarrow c = \frac{b^2}{a} \\ d &= \frac{c^2}{b} = \frac{b^4}{a^2} \cdot \frac{1}{b} = \frac{b^3}{a^2}, \quad e = \frac{d^2}{c} = \frac{b^6}{a^4} \cdot \frac{a}{b^2} = \frac{b^4}{a^3} \\ \therefore \frac{a}{e} &= \frac{a}{(b^4/a^3)} = \frac{a^4}{b^4} \end{aligned}$$

2. **Mean Proportional** If a, b and c are in continued proportion, then b is called the mean proportional of a and c .

i.e. $\frac{a}{b} = \frac{b}{c} \Rightarrow b^2 = ac \Rightarrow b = \sqrt{ac}$

EXAMPLE 6. What is the mean proportional between $(15 + \sqrt{200})$ and $(27 - \sqrt{648})$? ✎ 2012 I

- a. 4 b. $14\sqrt{7}$ c. $3\sqrt{5}$ d. $5\sqrt{3}$

Sol. c. Let mean proportional between $(15 + \sqrt{200})$ and $(27 - \sqrt{648})$ is x

$$\begin{aligned} \text{Then, } x &= \sqrt{(15 + \sqrt{200})(27 - \sqrt{648})} \\ &= \sqrt{(15 + 10\sqrt{2})(27 - 18\sqrt{2})} \\ &= \sqrt{5(3 + 2\sqrt{2}) \cdot 9(3 - 2\sqrt{2})} = \sqrt{45[(3)^2 - (2\sqrt{2})^2]} \\ &= \sqrt{45(9 - 8)} = \sqrt{45} = 3\sqrt{5} \end{aligned}$$

3. **Third Proportional** If $a : b :: b : c$, then c is called the 3rd proportional to a and b . Now, c will be calculated as below:

$$a : b :: b : c \Rightarrow a : b = b : c \Rightarrow c = \frac{b^2}{a}$$

EXAMPLE 7. The two numbers such that their mean proportional is 24 and the third proportional is 1536, are

- a. 3 and 98 b. 6 and 96 c. 4 and 92 d. None of these

Sol. b. Let a and b be the required numbers, then mean proportional between a and b is 24.

$$\therefore a : 24 = 24 : b \Rightarrow ab = 24^2, \quad ab = 576 \quad \dots(i)$$

and third proportional to a and b is 1536.

$$\Rightarrow a : b = b : 1536$$

$$\Rightarrow b^2 = 1536a \Rightarrow a = \frac{b^2}{1536} \quad \dots(ii)$$

On putting the value of a in Eq. (i), we get

$$\Rightarrow \frac{b^2}{1536} \cdot b = 576 \Rightarrow b^3 = 576 \times 1536$$

$$\Rightarrow b^3 = (96 \times 6) \times (96 \times 16) = 96^3 \Rightarrow b = 96$$

$$\therefore a = \frac{576}{b} = \frac{576}{96} = 6 \quad [\text{from Eq. (i)}]$$

Hence, the required numbers are 6 and 96.

4. **Fourth Proportional** If four numbers or quantities a, b, c and d are in proportion, then d is known as fourth proportional.

i.e. $a : b :: c : d$

EXAMPLE 8. The fourth proportional to

$$p^2 - pq + q^2, p^3 + q^3, p - q \text{ is}$$

- a. $p + q$ b. $p - q$ c. $p^2 + q^2$ d. $p^2 - q^2$

Sol. d. Let the fourth proportional be x .

$$\text{Then, } p^2 - pq + q^2 : p^3 + q^3 = p - q : x$$

$$\Rightarrow \frac{p^2 - pq + q^2}{p^3 + q^3} = \frac{p - q}{x} \Rightarrow x = \frac{(p - q)(p^3 + q^3)}{p^2 - pq + q^2}$$

$$\therefore x = \frac{(p - q)(p^3 + q^3)}{p^2 - pq + q^2} = (p - q)(p + q) = (p^2 - q^2)$$

Properties of Proportion

If four non zero quantities a, b, c and d are in proportion, then properties of proportion are as follows.

1. **Invertendo** If $a : b :: c : d$, then $b : a :: d : c$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$

2. **Alternendo** If $a : b :: c : d$, then $a : c :: b : d$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$

3. **Componendo** (Adding the Denominator)

If $a : b :: c : d$, then $(a + b) : b :: (c + d) : d$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a + b}{b} = \frac{c + d}{d}$

4. **Dividendo** (Subtracting the Denominator)

If $a : b :: c : d$, then $a : (a - b) :: c : (c - d)$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{a - b} = \frac{c}{c - d}$

5. **Componendo and Dividendo**

If $a : b :: c : d$, then $(a + b) : (a - b) :: (c + d) : (c - d)$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a + b}{a - b} = \frac{c + d}{c - d}$

EXAMPLE 9. If $(a - b) : (a + b) = 1 : 5$, then what is $(a^2 - b^2) : (a^2 + b^2)$ equal to?

- a. $\frac{13}{5}$ b. $\frac{4}{13}$ c. $\frac{5}{13}$ d. None of these

Sol. c. Given, $\frac{a - b}{a + b} = \frac{1}{5} \Rightarrow \frac{a + b}{a - b} = \frac{5}{1} \Rightarrow \frac{a + b + a - b}{a + b - a + b} = \frac{5 + 1}{5 - 1}$
(using componendo and dividendo rule)

$$\Rightarrow \frac{2a}{2b} = \frac{6}{4} = \frac{3}{2} \Rightarrow \frac{a}{b} = \frac{3}{2} \Rightarrow \frac{a^2}{b^2} = \frac{9}{4}$$

Again, using componendo and dividendo rule

$$\Rightarrow \frac{a^2 + b^2}{a^2 - b^2} = \frac{9 + 4}{9 - 4} = \frac{13}{5}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2 + b^2} = \frac{5}{13} \quad \therefore (a^2 - b^2) : (a^2 + b^2) = 5 : 13$$

MIXTURE OR ALLIGATION

If two or more quantities are mixed together in a certain ratio, then the product is called mixture.

Mean Price

Cost price of a unit quantity of the mixture is called the mean price.

Rule of Mixture or Alligation

It is the rule to find the ratio in which two or more ingredients are mixed or to be mixed.

If two quantities are mixed in the ratio $n_1:n_2$, then

$$\frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1}$$

where P_1, P_2 and P_m are cheaper price, dearer price and mean price respectively.

It can also be expressed as

$$\begin{array}{ccc} \text{Cheaper price} & & \text{Dearer price} \\ P_1 & & P_2 \\ & \swarrow \quad \searrow & \\ & \text{Mean price} & \\ & P_m & \\ & \swarrow \quad \searrow & \\ P_2 - P_m & & P_m - P_1 \\ \therefore \frac{\text{Cheaper quantity}}{\text{Dearer quantity}} = \frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1} \end{array}$$

Note $P_1 < P_m < P_2$

EXAMPLE 10. In what proportion, must wheat at ₹ 6.20 per kg be mixed with wheat at ₹ 7.20 per kg, so that the mixture be worth ₹ 6.50 per kg?

a. 6 : 4 b. 7 : 4 c. 7 : 3 d. 5 : 2

Sol. c. Given, cost price of cheaper quantity = ₹ 6.20 per kg

Cost price of dearer quantity = ₹ 7.20 per kg

Mean price = ₹ 6.50 per kg

According to the rule of alligation,

$$\begin{array}{ccc} \text{CP of cheaper} & & \text{CP of dearer} \\ \text{(620 paise)} & & \text{(720 paise)} \\ & \swarrow \quad \searrow & \\ & \text{Mean price} & \\ & \text{(650 paise)} & \\ & \swarrow \quad \searrow & \\ (720 - 650) & & (650 - 620) \\ = 70 \text{ paise} & & = 30 \text{ paise} \end{array}$$

$\therefore$ Required ratio = 70 : 30 = 7 : 3

VARIATIONS

When a change in a certain quantity leads to a certain change in another quantity then it is called variations.

Direct Variation

If x varies directly with y , then as x increases (decrease) y also increases (decreases)

i.e. $x \propto y$

$\Rightarrow x = ky$, where, k is the constant of variation.

Inverse Variation

If x varies inversely with y , then an increase is the value of x causes a decrease in the value of y .

i.e. $x \propto \frac{1}{y}$

$\Rightarrow x = \frac{k'}{y}$

where, k' is the constant of variation.

EXAMPLE 11. If a quantity y varies as the sum of three quantities of which the first varies as x , the second varies as $-x + x^2$, the third varies as $x^3 - x^2$, then what is y equal to?

- kx^3 , where k is a constant
- $kx + lx^2 + mx^3$, where k, l and m are constants
- kx^2 , where k is a constant
- kx , where k is a constant

Sol. b. Since, first term $\propto x$

$\Rightarrow$ First term = c_1x ,

$\Rightarrow$ Second term $\propto (-x + x^2)$

$\Rightarrow$ Second term = $c_2(-x + x^2)$

and third term $\propto (x^3 - x^2) \Rightarrow$ Third term = $c_3(x^3 - x^2)$

Also, $y \propto [c_1x + c_2(-x + x^2) + c_3(x^3 - x^2)]$ (given)

$\Rightarrow y = c_4[(c_1 - c_2)x + (c_2 - c_3)x^2 + c_3x^3]$

$$\begin{aligned} &= c_4(c_1 - c_2)x + (c_2 - c_3)c_4x^2 + c_3c_4x^3 \\ &= kx + lx^2 + mx^3 \end{aligned}$$

where, $k = c_4(c_1 - c_2)$,

$$l = (c_2 - c_3)c_4,$$

and $m = c_3c_4$

> PRACTICE EXERCISE

- The compounded ratio of $2a:6b$, $7a:49b$ and $3a:12b$ is
(a) $a:84b$ (b) $a^3:84b^3$ (c) $a:24b$ (d) $2a:3b$
- The subduplicate ratio of $16x^4:625y^6$ is
(a) $4x^2:25y^3$ (b) $4x^2:25y^2$ (c) $4x:25y$ (d) $8x^2:125y^3$
- If $x+5:3x+4$ is the duplicate ratio of $5:8$, then the value of x is
(a) 16 (b) 18 (c) 20 (d) 22
- If $x:6::5:3$, then the value of x is
(a) 8 (b) 10 (c) 12 (d) 13
- The third proportional to 9 and 12 is
(a) 12 (b) 14 (c) 16 (d) 18
- The mean proportional between $(2+\sqrt{3})$ and $(8-\sqrt{48})$ is
(a) 2 (b) 3 (c) 4 (d) 5
- The fourth proportional to 7, 11, 14 is
(a) 16 (b) 18 (c) 20 (d) 22
- In a ratio which is equal to $7:8$, if the antecedent is 35, then the consequent is
(a) 30 (b) 32 (c) 36 (d) 40
- If $x:y=1:3$, $y:z=5:k$, $z:t=2:5$ and $t:x=3:4$, then what is the value of k ?
(a) $1/2$ (b) $1/3$ (c) 2 (d) 3
- If the ratio of x to y is 25 times the ratio of y to x , then what is the ratio of x to y ?
(a) $1:5$ (b) $5:1$ (c) $25:1$ (d) $1:25$
- If $p\%$ of ₹ x is equal to t times $q\%$ of ₹ y , then what is the ratio of x to y ?
(a) $pt:q$ (b) $p:qt$ (c) $qt:p$ (d) $q:pt$
- If $a:b=2:3$ and $x:y=3:4$, then the value of $\frac{4ay-3bx}{5ax-2by}$ is
(a) $\frac{5}{3}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) $\frac{5}{4}$
- If $\frac{x^3+3x}{3x^2+1}=\frac{341}{91}$, then the value of x is
(a) 16 (b) 14 (c) 13 (d) 11
- What is the number which has to be added to each term of the ratio $49:68$, so that it becomes $3:4$?
(a) 3 (b) 5 (c) 8 (d) 9
- Divide 1870 into three parts in such a way that half of the first part, one-third of the second part and one-sixth of the third part are equal. Then, the third part is
(a) 340 (b) 510 (c) 1020 (d) 1320
- Out of the ratios $7:20$, $13:25$, $17:30$ and $11:15$, the smallest one is
(a) $11:15$ (b) $7:20$ (c) $11:16$ (d) $17:30$
- What is the ratio of the numbers of 0.5 of a number is equal to 0.07 of another?
(a) $50:7$ (b) $5:7$ (c) $1:14$ (d) $7:50$
- What is the ratio whose terms differ by 40 and the measure of which is $\frac{2}{7}$?
(a) $14:56$ (b) $15:56$ (c) $16:56$ (d) $16:72$
- If $P:Q=\frac{3}{5}:\frac{5}{7}$ and $Q:R=\frac{3}{4}:\frac{2}{5}$, then what is $P:Q:R$ equal to?
(a) $\frac{3}{5}:\frac{5}{7}:\frac{2}{5}$ (b) $\frac{9}{20}:\frac{15}{28}:\frac{2}{7}$ (c) $\frac{3}{5}:\frac{3}{4}:\frac{2}{5}$ (d) $\frac{3}{5}:\frac{5}{7}:\frac{3}{4}$
- Two numbers are in the ratio $3:5$. If 9 is subtracted from each number, then they are in the ratio of $12:23$. What is the second number?
(a) 44 (b) 55 (c) 66 (d) 77
- The mean proportional between two numbers is 28 and their third proportional to them is 224. The two numbers are
(a) 7 and 112 (b) 14 and 56 (c) 28 and 28 (d) 21 and 36
- ₹ 770 have been divided among A, B, C in such a way that A receives $\frac{2}{9}$ th of what B and C together receive. What is A 's share?
(a) ₹ 154 (b) ₹ 140 (c) ₹ 250 (d) ₹ 254
- x varies directly as y and inversely as square of z . When $y=4$ and $z=14$, then $x=10$. If $y=16$ and $z=7$, what is x ?
(a) 180 (b) 160 (c) 154 (d) 140
- If $x=\frac{\sqrt{m+3n}+\sqrt{m-3n}}{\sqrt{m+3n}-\sqrt{m-3n}}$, then
(a) $3nx^2-2mx+3n=0$ (b) $2nx^2-2mx+3n=0$
(c) $3nx^2-2mx-3n=0$ (d) $3nx^2+3mx+3n=0$
- If $p+r=2q$ and $\frac{1}{q}+\frac{1}{s}=\frac{2}{r}$, then
(a) $p:s=r:q$ (b) $p^2:r^2=q:s$
(c) $p:q=r:s$ (d) $p^2=2r+s$